Chu Wei Chang

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Education

National Cheng Kung University (NCKU) Tainan, Taiwan
M.S. in Medical Informatics Sep. 2023 – Present
B.S. in Computer Science and Information Engineering Sep. 2019 – Jun. 2023

TOEFL iBT: 107 (Reading 29, Listening 30, Speaking 24, Writing 24)

Skills

Languages: C, Python, Rust, Go , SQL, TypeScript/JavaScript
Systems: Linux kernel, syscall interception (ptrace, seccomp-BPF), AddressSanitizer (ASan)
ML & Vision: PyTorch, TensorFlow/TFLite, LLM fine-tuning (LoRA, quantization)
Web & Data: Angular, ConnectRPC, PostgreSQL, CockroachDB

Work Experience

National Cheng Kung University | Teaching Assistant, Algorithms Feb. 2026 – Jun. 2026
- Designed 7 homework assignments plus the midterm and final for a 180-student undergraduate course.
- Graded all homework and exams and gave students feedback on their solutions.

MayaInfo Co., Ltd. | Software Engineer Dec. 2024 – Feb. 2025
- Designed the user workflow for a client's real-estate pricing-catalog web app from their requirements.
- Designed the database schema (ERD) and RESTful APIs for the website's backend.
- Prototyped the core page layouts in Figma and handed them off to the design team.

Aco Healthcare Co. | Computer Vision Engineer *(Part-Time, 20 hrs/wk)* Jun. 2024 – Dec. 2024
- Led a project to automatically switch imaging modes on an ultrasound probe.
- Trained an EfficientNetV2 classifier to recognize 8 scanning scenes across 10+ tissue types, reaching 90% top-1 accuracy.
- Designed a mode-switching policy that combines sampling and filtering to select the optimal mode.

Aco Healthcare Co. | Intern Engineer *(Part-Time, 7 hrs/wk)* Mar. 2023 – Jun. 2023
- Extended our capstone intima-thickness model and integrated it into the company's platform.
- Raised Dice from 64.0% to 82.2% by denoising and cropping to the carotid walls (detected at 96.2%).

Projects

kbox | Rootless Linux Sandbox (In-Process Kernel + seccomp)
C, Linux Kernel Library (LKL), seccomp, AddressSanitizer (ASan), GitHub Actions
- Traced ASan-reported stack overflows to undersized custom ioctl structs that broke the 64-bit ABI (PR #34).
- Replaced them with the kernel's `struct ifreq` and `rtentry`, adding `_Static_assert` size guards.
- Fixed reverse-map entries stuck in MULTI state after `dup2`, which forced O(n) fallback scans (PR #68).
- Fixed `sendfile` writes bypassing the shadow FD, so close-time sync missed the data (PR #65).
- Proposed and built a `make check-perf` CI target; lookups held at 2–4 ns from 64 to 16,384 entries (PR #70).
- Maintain a fork carrying these fixes while PRs #65, #68 and #70 await upstream review.

JetSeek | Real-Time Object-Searching Robot
Python, asyncio, YOLOv8, TCP Sockets, NVIDIA JetBot
- Built a PC-driven robot: a JetBot streams frames over TCP and the PC runs YOLOv8 on 11 target classes.
- Designed an asyncio pub/sub architecture that decouples frame intake, detection, search and motor control.
- Implemented a scan-and-relocate search (six 60° scans, then a random move) with turn calibration.
- Wrote a 10 Hz `Commander` that picks search or approach commands based on whether the target is visible.

Carotid Artery Screening by Wireless Handheld Ultrasound | Capstone Project, Team of 4 2023
Python, TensorFlow Lite, C++, Java, Android NDK
- Built a screening system that segments the carotid intima with U-Net and measures its thickness.
- Ran the model on Android tablet in C++ (NDK) with TensorFlow Lite, paired with a wireless probe.
- Won 2nd place (top 4 of 64 teams) at the NCKU CSIE Project Exhibition.